

Burst-suppression — labeling clinical EEG reports, and comparison with Harvard

We labeled every report for whether it explicitly describes a **burst-suppression** (or suppression-burst) pattern. Three-way call: the report describes it (present), the report says it is absent (explicitly_absent), or it isn't mentioned (not_mentioned). Vasily's suggested prompt, used verbatim, grammar-constrained so the answer is exactly one of those three. MedGemma-27B (Q4_K_S), temperature 0.

It ran on Fraser Health (45,545 reports) and three Harvard hospitals — MGH, BWH, BCH (~100,600; BIDMC has no text). About 1.3% of Harvard reports are longer than the context window and returned nothing; they are excluded.

The matching Harvard variable

This is the one of the three background fields that Harvard also has a column for: **bs**. So here we can compare head-to-head. A short reminder of how Harvard's column works (full detail in reports/harvard_llm_labeling.md): a Harvard cell is not a yes/no — it is a *provenance* string saying how the finding was established (from the report text, from the recording's structured annotations, or expert-verified) or it is empty. Empty means "not recorded", not "absent". We compare three ways:

- **vs all** — against everything Harvard marked (any non-empty cell).
- **text-only** — against only what Harvard took from the report text, which is the fair like-for-like since we also read only the text.
- **recall on verified** — of the cases an expert had confirmed as burst-suppression, how many we also called present.

What we found

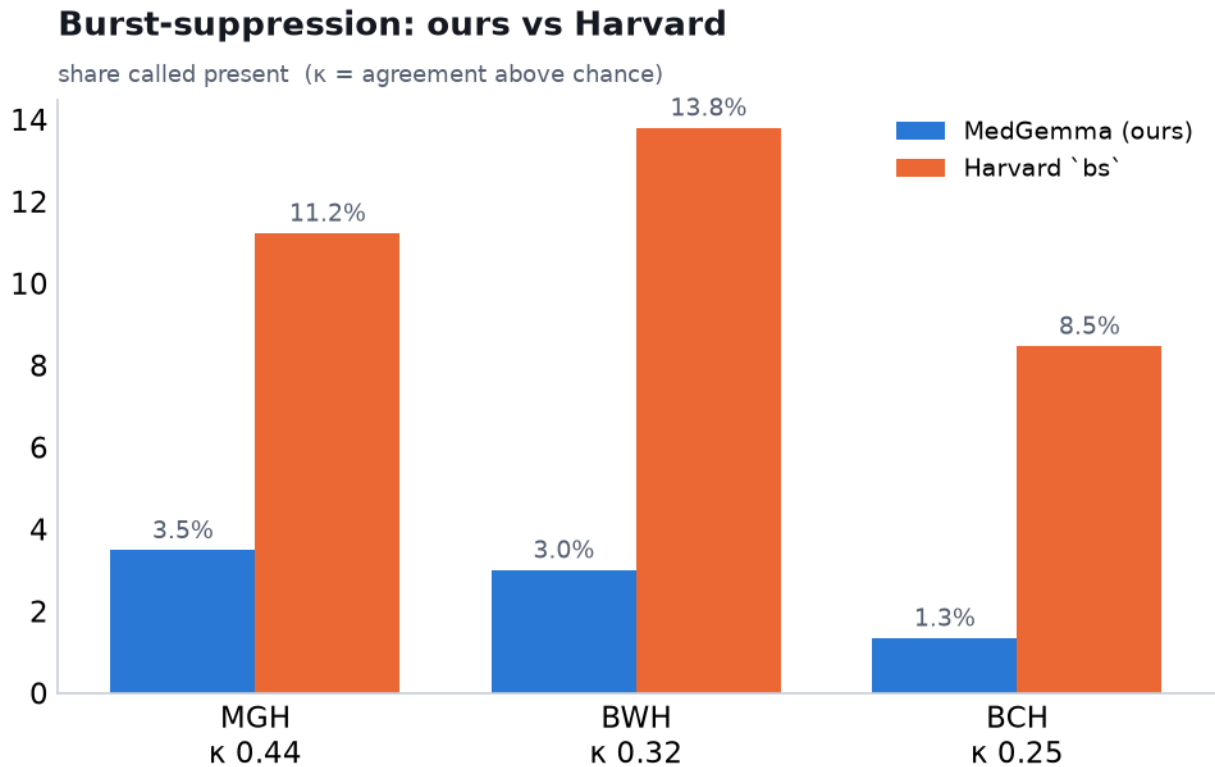


Figure 1: Burst-suppression: ours vs Harvard bs

	ours present	Harvard bs present	agreement	κ	κ (text-only)	our recall on verified
MGH	3.5%	11.2%	92%	0.44	0.60	65% (231)
BWH	3.0%	13.8%	89%	0.32	0.50	61% (184)
BCH	1.3%	8.5%	93%	0.25	0.29	— (0 verified)

A note on the metrics. *Agreement* is how often the two sides make the same present/not call. It looks high (89–93%), but that is mostly because the finding is rare — both sides say “no” on almost every report, so they agree by default. κ (*kappa*) is agreement after subtracting what chance alone would give (0 = chance, 1 = perfect); it is the honest number, and here it is fair-to-moderate. κ *text-only* compares against only the labels Harvard took from the report text. *Recall on verified* is how many expert-confirmed cases we caught (positives only; BCH has no verified cases).

Two things stand out.

Harvard’s bs is marked far more often than we mark it — 11% of MGH reports, 14% at BWH, against our 1–3.5%. That gap is not the report narratives disagreeing with us. It’s Harvard’s annotation channel: when we compare only against the labels Harvard took from the report *text*, κ jumps to 0.60 at MGH and 0.50 at BWH. So most of the extra bs comes from the recordings’ structured annotations, which flag burst-suppression on many recordings whose written report doesn’t describe it — and which we, reading only the text, never see.

BCH is the exception. There Harvard’s bs is itself mostly text-derived (about 90% of its marks carry a report tag), so restricting to the text barely changes anything (κ 0.25 $\rightarrow$ 0.29). The gap at BCH — their 8.5% against our 1.3% — is therefore a genuine disagreement on the reports themselves: we read burst-suppression out of those narratives less often than Harvard’s model did, not an annotation artifact.

On the expert-verified cases we catch about 61–65%. Part of the miss is the same channel issue (some verified labels are annotation-derived and never made it into the narrative), so this is a floor rather than a clean recall figure.

The full three-way distribution of our own labels:

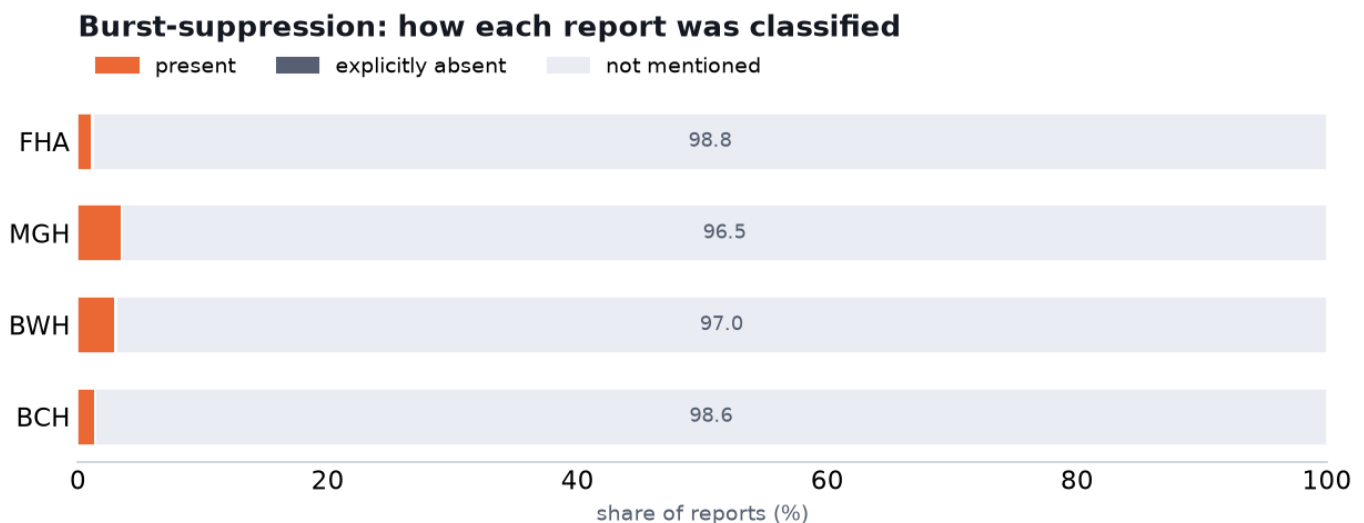


Figure 2: Burst-suppression — how each report was classified

dataset	present	explicitly absent	not mentioned	reports
Fraser Health	1.1%	0.1%	98.8%	45,545
MGH	3.5%	0.1%	96.5%	54,854
BWH	3.0%	0.0%	97.0%	20,716

dataset	present	explicitly absent	not mentioned	reports
BCH	1.3%	0.0%	98.6%	25,022

Reports almost never say burst-suppression is *absent* — when the pattern isn’t there, the report simply doesn’t bring it up. So `explicitly_absent` is near zero and the split is really present-vs-not-mentioned.

Caveats

- Harvard’s `bs` value mixes report-derived, annotation-derived and expert-verified flags, and a blank cell mixes “absent” with “not mentioned” and “not annotated”. We keep those apart on our side but collapse to present-vs-not to compare.
- The verified check is positives-only (there is no verified “absent”) and has no BCH data, so it measures recall, not precision.
- About 1.3% of Harvard reports were too long for the context window and returned no label; they are excluded.

Reproduce

```
# labeling (on the cluster)
bash slurm/submit_background.sh           # Fraser Health
bash slurm/submit_background_harvard.sh   # MGH / BWH / BCH
# comparison + charts
python -m analysis.background_vs_harvard  # results/harvard/burst_vs_harvard.json
python -m analysis.background_chart        # figures/background_burst.png
python -m analysis.background_field_charts # figures/dist_burst_suppression.png
```

Labeler and prompt: `src/cpu/background.py` (field `burst_suppression`). How Harvard’s `bs` column is coded: `reports/harvard_llm_labeling.md`.